CS 2614: Computer Organization

Solutions: Homework # 2

1.

3-7. Show the value of all bits of a 12-bit register that hold the number equivalent to decimal 215 in (a) binary; (b) binary-coded octal; (c) binary-coded hexadecimal; (d) binary-coded decimal (BCD).

Solution:

$$(215)_{10} = 128+64+16+7 = (11010111)_2$$

(a) 000011010111 Binary

(b) 000 011 010 111 Binary coded octal
  0 3 2 7

(c) 0000 1101 0111 Binary coded hexadecimal
  0 D 7

(d) 0010 0001 0101 Binary coded decimal
  2 1 5

2.

3-9. Write your name in ASCII using an 8-bit code with the leftmost bit always 0. Include a space between names and a period after a middle initial.

Solution:

Here the students should write their names. For example, for 'MAN':

01001101 01000001 01001110

3.

3-12. Obtain the 10's complement of the following six-digit decimal numbers: 123900; 090657; 100000; and 000000.

Solution:

876100; 909343; 900000; 000000

4.

Formatted: Left

3-15. Perform the subtraction with the following unsigned binary numbers by taking the 2's complement of the subtrahend.

a. 11010 − 10000 b. 11010 − 1101

c. 100 − 110000 d. 1010100 − 1010100

Solution:

a.

0 1 0 0 0 0

2's complement 1 1 0 0 0 0

Now, 0 1 1 0 1 0

\+ 1 1 0 0 0 0

$\boxed{\cancel{1}}$ 0 0 1 0 1 0

Verification: 26 – 16 = 10

b.

0 0 1 1 0 1

2's complement 1 1 0 0 1 1

Now, 0 1 1 0 1 0

1 1 0 0 1 1

$\boxed{\cancel{1}}$ 0 0 1 1 0 1

Verification: 26 – 13 = 13

c.

0 1 1 0 0 0 0

2's complement 1 0 1 0 0 0 0

Now 0 0 0 0 1 0 0

1 0 1 0 0 0 0

1 0 1 0 1 0 0

2's complement 0 1 0 1 1 0 0

Verification: 4 – 48 = -44

d.

0 1 0 1 0 1 0 0

2's complement 1 0 1 0 1 1 0 0

Now, 0 1 0 1 0 1 0 0

1 0 1 0 1 1 0 0

$\boxed{\cancel{1}}$ 0 0 0 0 0 0 0 0

Verification: 84 – 84 = 0

5. Convert the following numbers to gray code:
   a. 01001_2

b. 43_{10}
c. $3A_{16}$
d. 11111_2
e. 55_8
f. 101001_2
g. 110101_2

Solution:

a. 01001_2

01101

b. 43_{10}

43_{10}	
21	1
10	1
5	0
2	1
1	0
0	1

$= (101011)_2$

Ans. 111110

c. $3A_{16}$

$= (00111010)_2$ Ans. 00100111

d. 11111_2

10000

e. 55_8

$(\underbrace{5}\underbrace{5})_8 = (\underbrace{101}\underbrace{101})_2$

Ans/ 111011

f. 101001_2

111101

g. 110101_2

101111

6. Convert the following numbers represented in gray code to the bases shown:
   a. 1001 to binary
   b. 10011 to hexadecimal
   c. 10001 to octal
   d. 11111 to decimal

Solution:

a. 1001 to binary

1110

b. 10011 to hexadecimal

$(11101)_2 = (1D)_{16}$

c. 10001 to octal

$(11110)_2 = (36)_8$

d. 11111 to decimal

$(10101)_2$

$= (1\times 2^4 + 0 + 1\times 2^2 + 0 + 1\times 2^0)_{10}$

$= (16+4+1)_{10}$

$= (21)_{10}$

7. Find the ~~two's complement~~ binary number ~~representation~~ of 42 (decimal) using 8-bit two's complement representation ~~numbers~~.

Solution:

$(42)_{10}$

21	0
10	1
5	0
2	1
1	0
6	1

$(42)_{10} = (101010)_2$

2's Complement

$= 0101010$

8. Find the ~~two's complement~~ binary number ~~representation~~ of -42 (decimal) using 8-bit two's complement representation ~~numbers~~.

Solution:

00101010

11010110

9. Consider this two's complement binary number: 10011110. What is its decimal value?

Solution:

(1)0011110

$(-)1100010 = (-)64 + 32 + 2 = -98$

10. Find the 8-bit two's complement binary number representation of 137 (decimal). Briefly describe what went wrong, and what that implies about the limits of this representation scheme?

Solution:

Using 8 bits, up to 127 can be written (signed). So, 9 bits require.

28 = 256

For unsigned, 0 – 256

For signed, maximum +127 can be written.

So, for 137, 9 bits require.

11. Represent the following numbers using 8-bit 2’s complement representation
    a. -47_{10}
    b. -123_{8}
    c. $5C_{16}$

Solution:

a.

$$(-47)_{10} = \underbrace{1}_{(-)}\underbrace{0101111}_{(47)}$$

2's Complement = 11010001

b.

$$(-123)_8 = (11010011)_2$$

2's Complement = 10101101

c.

$$(5C)_{16} = (01011100)_2 = \text{2's Complement}$$

2's Complement = 01011100

12. Perform the following arithmetic operations on the two's complement numbers given below. Indicate whether an overflow occurs or not. The first bit in each number is a sign bit.
    a. 10110 + 11001
    b. 11001 + 11000

Solution:

a. Overflow

Carry: 1 0

$$\begin{array}{r} 1\,0\,1\,1\,0 \\ 1\,1\,0\,0\,1 \\ \hline \boxed{1}\,0\,1\,1\,1\,1 \end{array}$$

b. No Overflow

Carry: 1 1

$$\begin{array}{r} 1\,1\,0\,0\,1 \\ 1\,1\,0\,0\,0 \\ \hline \boxed{1}\,1\,0\,0\,0\,1 \end{array}$$

13. Convert the following 2's complement binary numbers to decimal.
    a. 0110
    b. 1101
    c. 0110 1111
    d. 1101 1011 0001 1100

Solution:

a.

$$\overset{2^2\ 2^1\ 2^0}{0110} = (6)_{10}$$

(+)

b.

$$1101 = -(0011) = (-3)_{10}$$
$$= 1011$$

c.

$$\overset{6\ 5\ \ 3\ 2\ 1\ 0}{01101111} = 64+32+8+4+2+1$$
$$= 111$$

d.

$$1101101100011100$$
$$= 1010010011100100$$
$$= -(8192+1024+128+64+32+4)$$
$$= -9444$$

14. The following binary numbers are 4-bit 2's complement binary numbers. Which of the following operations generate overflow? Justify your answers by translating the operands and results into decimal.
    a. 0011 + 1100
    b. 0111 + 1111
    c. 1110 + 1000
    d. 0110 + 0010

Solution:

a. ***0011 + 1100***

Adding in decimal equivalent:

$3 + -[NOT(1100)+1] = 3 -(0011+1)_2 = (3-4)_{10} = -1_{10}$

Adding in 2's complement:

$0011 + 1100 = (1111)_2 = -[NOT(1111)+1] = -1_{10}$ **(No Overflow)**

b. ***0111 + 1111***

Adding in decimal equivalent:

$7 + -[NOT(1111)+1] = 7 -1 = 6_{10}$

Adding in 2's complement:

$0111 + 1111 = (0110)_2 = 6_{10}$ **(No Overflow)**

c. ***1110 + 1000***

Adding in decimal equivalent:

$-[NOT(1110) + 1] - [NOT(1000) + 1] = -(10)_2 - (1000)_2 = -2 + -8 = -10$

Adding in 2's complement:

$1110 + 1000 = 0110 = 6_{10}$ **(Overflow!)**

d. 0110 + 0010

Adding in decimal equivalent:

6 + 2 = 8

Adding in 2's complement:

1 000 = **-** [NOT(1000) + 1] = **-**(0111+1) = **-**(1000) = **-**8 **(Overflow)**

15. Consider the following numbers as 4-bit signed binary number. Indicate whether the result will generate any overflow, or not?
    a. 1111 + 1000
    b. 1100 + 0100
    c. 0100 + 0011
    d. 0001 + 0111

Solution:

a.

1111 + 1000

1 1 1 1

1's complement: 0 0 0 0

+1

2's complement - 0 0 0 1 = $(-1)_{10}$

1 0 0 0

1's complement: 0 1 1 1

+ 1

2's complement: -1 0 0 0 $= (-8)_{10}$

$$\begin{array}{lcl} 1111 & = & (-1)_{10} \\ 1000 & = & +(-8)_{10} \\ \hline \boxed{1}0111 & & (-9)_{10} \end{array}$$

↓

7

overflow.

b.
1100 + 0100

1 1 0 0

1's complement: 0011

+1

2's complement: -0100

$= (-4)_{10}$

$$
\begin{array}{r l}
1100 & = (-4)_{10} \\
0100 & = (4)_{10} \\
\hline
\boxed{1}\,0000 & \quad 0
\end{array}
$$

No overflow.

c.

100 011

$4 + 3 = (7)_{10}$

$0100 + 0011 = (0111)_2 = (7)_{10}$ No overflow.

d.

0001 + 0111

$1 + 7 = (8)_{10}$

$0001 + 0111 = (1000)_2 = (-8)_{10}$ overflow

16. Consider the following numbers. Indicate whether the addition operation will generate any overflow. Show the results, considering the numbers as (i) signed and (ii) unsigned
    a. $12_{10} + 8_{10}$ (Represent the numbers in 5-bit binary)
    b. $A_{16} + 24_{16}$ (Represent the numbers in 6-bit binary)

Solution:

a.

$12_{10} + 8_{10}$ (Represent the numbers in 5-bit binary)

(i)

$= (2^3 + 2^2 + 0 + 0 + 2^3 + 0 + 0 + 0)_{10}$

$= (01100 + 01000)_2$

$= \boxed{1}0100 = (-4)_{10}$ Overflow.

$$\begin{array}{r} 12 \\ +\ 8 \\ \hline (20)_{10} \end{array}$$

(ii)

$$\begin{array}{r} 01100 \\ 01000 \\ \hline 10100 \end{array}$$

$= (20)_{10} (< 32)$ No overflow.

b.

$A_{16} + 24_{16}$ (Represent the numbers in 6-bit binary)

(i)

$100100 = -(NOT(100100)+1) =$

$(-)\ 011011$

$\underline{000001}$

$(-)011100 = -28_{10}$

$001010 \quad 10_{10}$

$+\underline{100100} \quad \underline{-28_{10}}$

$101110 \quad -18_{10}$

$101110 = -(NOT(101110)+1) =$

$(-)\ 010001$

$+\ \underline{000001}$

$(-)010010 = -18_{10}$

(ii)

$001010 = 10_{10}$

$+\underline{100100} \quad \underline{+36_{10}}$

$101110 \quad 46 < 64_{10}$

$= 46_{10}$ No over flow

17. What is the 32-bit IEEE 754 floating point representation for 3.5?
Solution:

In binary fixed point, 3.5 is 11.1

Normalized, 11.1 is 1.11×2^{1}

IEEE 754 sign field is 0 (for positive)

IEEE 754 exponent field is 127+1 or 10000000 in binary

IEEE 754 fraction field is 127+1 or 11000000000000000000000

IEEE 754 representation is 01000000011000000000000000000000 or 40600000 (hex)

18. If $(42510000)_{16}$ is the 32-bit IEEE 754 representation of a floating-point number, what is the floating-point number?

Solution:

42510000 is 01000010010100010000000000000000

For IEEE 754, can be divided into 0.10000100.10100010000000000000000

0 means number is positive

10000100 means exponent field is 132 and binary exponent is 132-127 or 5 so exponent multiplier is 2^5 or 32

10100010000000000000000 means fraction is 1.10100010000000000000000 or 1+1/2+1/8+1/128 or 1+64/128+16/128+1/128 or 1+81/128 or 1.6328125

Result is +32×1.6328125 or 52.25

19. Show the IEEE 754 binary representation for the following floating-point numbers
    a. 356.75
    b. -11.5

Solution:

a. 356.75 = 101100100.11 = 1.0110010011 * 2^8

Sign = 0

Fraction = 0110010011

Single exponent = 8 + 127 = 135

Double exponent = 8 + 1023 = 1031

Single precision:

1	8	23
0	1000 0111	011 0010 0110 0000 0000 0000

Double precision:

1	11	52
0	100 0000 0111	0110 0100 1100 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000

b. 11.5 = 1011.1 = 1.0111 * 2^3

Sign = 1

Fraction = 0111

Single exponent = 3 + 127 = 130

Double exponent = 3 + 1023 = 1026

Single precision:

1	8	23
1	1000 0010	011 1000 0000 0000 0000 0000

Double precision:

1	11	52
1	100 0000 0010	0111 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000

20. Convert the following numbers represented in IEEE 754 floating point representation to their equivalent decimal numbers.
    a. 0 111 1111 1 000 0100 0100 1111 0000 0000
    b. 0 101 1100 1 010 0100 0100 0000 0000 0000
    c. 1 000 1101 1 011 0110 1000 0000 0000 0000
    d. 1 000 0000 0 011 1011 0000 0000 0000 0000

Solution:

a) 0 111 1111 1 000 0100 0100 1111 0000 0000

Sign = 0
Exponent = 11111111
Fraction = 000 0100 0100 1111 0000 0000
NAN

b) 0 101 1100 1 010 0100 0100 0000 0000 0000

Sign = 0
Exponent = 1011 1001 = 185
Fraction = 01001000100000000000000
$(-1)^0 (1+0.010010001)2^{185-127} = (1+2^{-2}+2^{-5}+2^{-9})* 2^{58} = 3.6986 \times 10^{17}$

c) 1 000 1101 1 011 0110 1000 0000 0000 0000

Sign = 1

Exponent = 0001 1011 = 27

Fraction = 01101101000000000000000

$(-1)^1 (1+0.01101101)2^{27-127} = (1+2^{-2}+2^{-3}+2^{-5}+2^{-6}+2^{-8})* 2^{-100} = -1.1247 \times 10^{-30}$

d) 1 000 0000 0 011 1011 0000 0000 0000 0000

Sign = 1

Exponent = 0000 0000 → denormalized

Fraction = 01110110000000000000000

$(-1)^1 (0.0111011)2^{0-127+1} = (2^{-2}+2^{-3}+2^{-4}+2^{-6}+2^{-7})* 2^{-126} = -5.4183 \times 10^{-39}$

21. The floating-point representations of X and Y are given below. Perform X + Y. ('+' symbol means addition here not 'OR').

X = 1 011 1011 1 100 1000 0000 0000 0000 0000

Y = 0 100 0100 1 001 0100 1000 0000 0000 0000

Solution:

x:

Sign = 1

Exponent = 0111 0111 = 119

Fraction = 100 1000 0000 0000 0000 0000

$x = (-1)^1 (1+0.1001)2^{119-127} = -1.1001 \times 2^{-8}$

y:

Sign = 0

Exponent = 1000 1001 = 137

Fraction = 001 0100 1000 0000 0000 0000

$y = (-1)^0 (1+0.00101001)2^{137-127} = 1.00101001 \times 2^{10}$

a) x+y

Step 1: lesser exponent -> larger exponent

$x = -1.1001 \times 2^{-8} = -0.0000\ 0000\ 0000\ 0000\ 0110\ 01 \times 2^{10}$

Step 2: add the significands:

-0.0000 0000 0000 0000 0110 010 + 1.0010 1001

$$
\begin{array}{r}
1.0010\ 1001\ 0000\ 0000\ 0000\ 000 \\
-\ 0.0000\ 0000\ 0000\ 0000\ 0110\ 010 \\
\hline
=\ 1.0010\ 1000\ 1111\ 1111\ 1001\ 110
\end{array}
$$

Step 3: Normalize, no change

Step 4: Round, no change

Sign = 0

Exponent = 10+127 = 137 = 10001101

Fraction = 0010 1000 1111 1111 1001 110

0 100 0100 1 0010 1000 1111 1111 1001 110

0	10001001	0010 1000 1111 1111 1001 110

22. Floating-Point Addition: add the following binary numbers
    a. $1.000 * 2^{-1} + 1.110 * 2^{-2}$
    b. $1.000 * 2^{-1} + -1.110 * 2^{-2}$ or, $1.000 * 2^{-1} + (-1.110 * 2^{-2})$
    c. $1.010 * 2^{-2} + 1.110 * 2^{-3}$

Solution:

a. $1.000 * 2^{-1} + 1.110 * 2^{-2}$

$= 1.000 \times 2^{-1} + 0.111 \times 2^{-1}$

$= 1.111 \times 2^{-1}$

b. $1.000 * 2^{-1} + -1.110 * 2^{-2}$ or, $1.000 * 2^{-1} + (-1.110 * 2^{-2})$

$= 1.000 \times 2^{-1} + -0.111 \times 2^{-1}$

$= 0.001 \times 2^{-1}$

$= 1 \times 2^{-4}$

or, 1.000×2^{-4}

c. $1.010 * 2^{-2} + 1.110 * 2^{-3}$

$= 1.010 \times 2^{-2} + 0.111 \times 2^{-2}$

$= 10.001 \times 2^{-2}$

$= 1.0001 \times 2^{-1}$

23. Floating-Point Addition: add the following numbers (converting into binaries)
    a. $5.75_{10} + 2.25_{10}$
    b. $3.5_{10} + 0.25_{10}$

Solution:

a.

5	
2	1
1	0
0	1

$(5)_{10} = 101$

0.75×2

$= \boxed{1}.50 \times 2$

$\;\;\boxed{1}.00$

$(0.75)_{10} = (0.11)_2$

$5.75_{10} + 2.25_{10} = 101.11 + 10.01$

$= 1000.00$

$= 1 \times 2^3$

b. $(2)_{10} = (10)_2$

0.25×2

$= 0.50 \times 2$

$= 1.00$

$(0.25)_{10} = (0.01)_2$

$3.5_{10} + 0.25_{10} = 11.1 + 0.01$

$= 11.11$

$= 1.111 \times 2^1$

24. Write the parity bit (both even and odd) for the following binary numbers:
    a. 1100
    b. 1000
    c. 0000
    d. 1101

Solution:

	P(odd)	P(even)
a. 1100	1	0
b. 1000	0	1
c. 0000	1	0
d. 1101	0	1